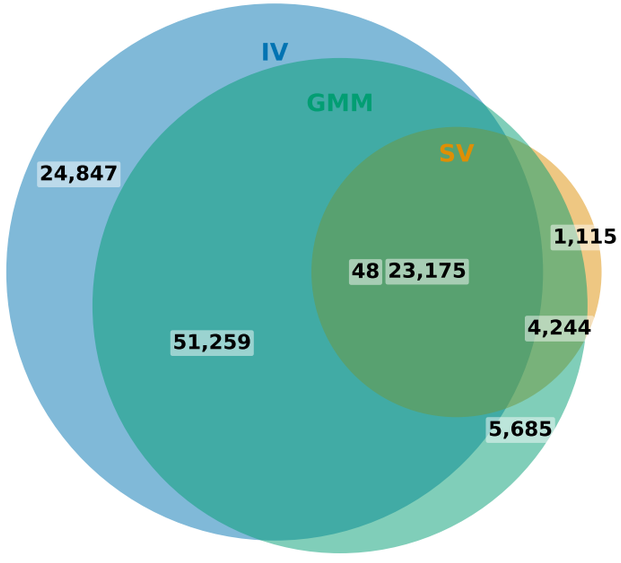
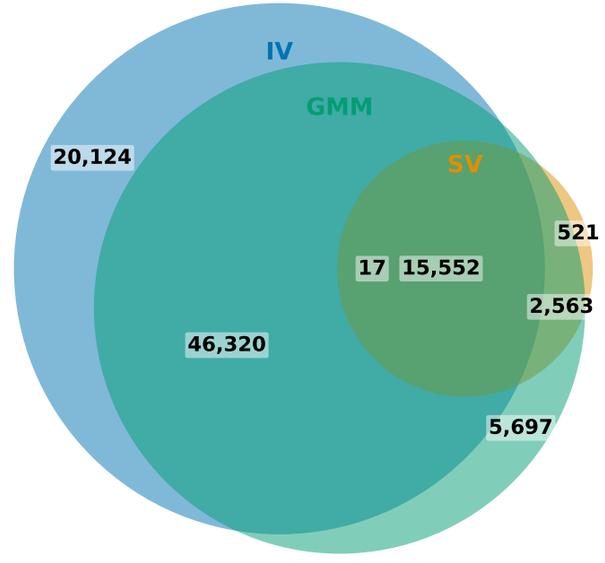
Real-data significance overlap and GO:BP enrichment

(a) Wald: BH $q \leq 0.05$

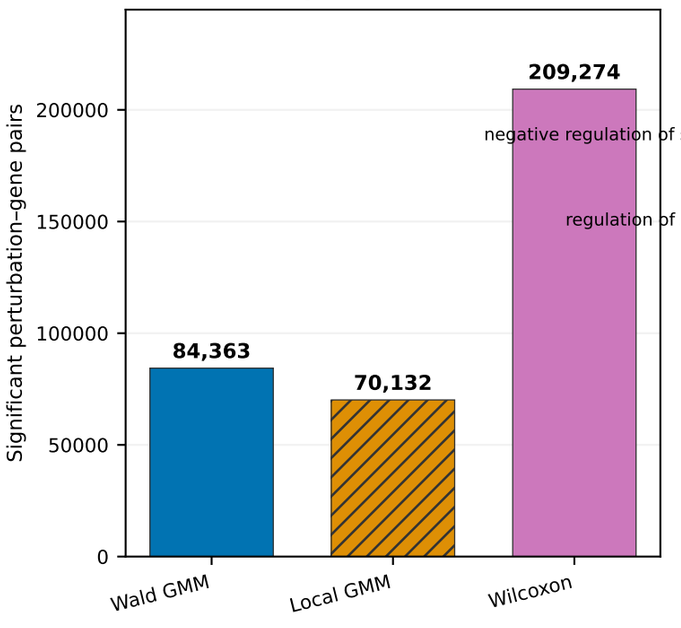


(b) Local robust: BH $q \leq 0.05$

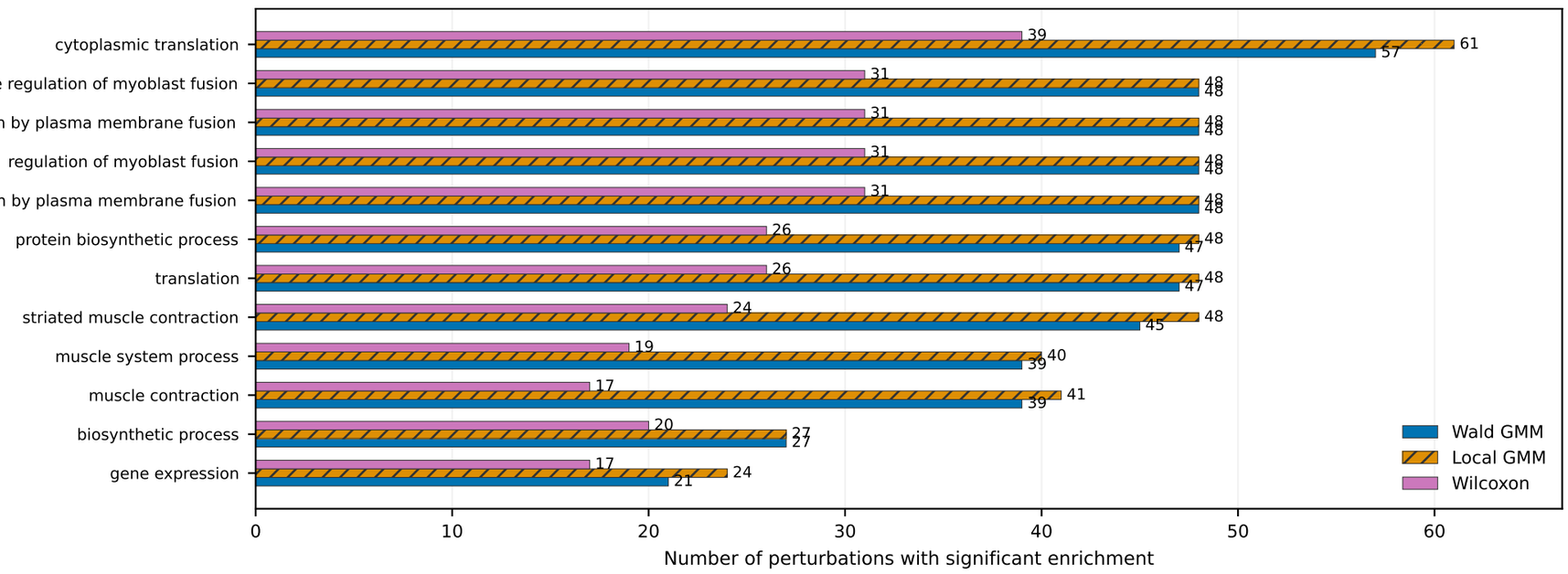


Circle areas and inclusive pairwise overlaps are proportional to counts; region labels are exact.

(c) DE genes used for ORA



(d) Most recurrent significant GO:BP terms
GMM Wald 1,403, local 1,655; Wilcoxon 894



Median selected genes per perturbation: Wald GMM 309, local GMM 229, Wilcoxon 1180. GMM selects fewer genes than Wilcoxon in 195/213 (Wald) and 203/213 (local) perturbations. GO background: 2,000 tested outcomes.